Neel Gali

Senior Software Engineer, AI/ML

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Summary

Senior Software Engineer with 5+ years building **reliable distributed backend systems** and **production agentic AI platforms** across **Python**, **TypeScript**, and **Java**, with hands-on experience in **LangGraph**, **Pydantic AI**, **LangChain**, **CrewAI**, **RAG**, balancing **reliability and scale** with experimentation for 10,000+ enterprise users on **AWS** and **Azure**.

Professional Experience

Senior Software Engineer, AI/ML *Jul 2025 - Present*

Epik Solutions | Enterprise Security Platform

- Architected **LangGraph** stateful agent workflows with persistent **memory systems** (checkpointer-backed conversation state and cross-session context), multi-turn security agents reduced analyst triage time by 40%.
- Built **human-in-the-loop** review workflows using LangGraph interrupt nodes, allowing security analysts to approve, reject, or redirect agent actions mid-execution before high-stakes remediation steps were taken.
- Designed **Pydantic AI** agent tool schemas with strict type-safe structured outputs, eliminating unstructured LLM response parsing and reducing downstream processing errors by 45% across all production agent pipelines.
- Built **RAG** pipelines over enterprise security corpora using **Pinecone** vector database and **OpenAI** embeddings with hybrid semantic + keyword retrieval, improving context-relevance accuracy by 38% over baseline.
- Implemented **MLOps** practices for LLM services: automated evaluation pipelines tracking faithfulness, groundedness, and latency regressions, with **GitHub Actions** CI/CD gating deployments on quality thresholds.
- Built **FastAPI** and **Node.js GraphQL/REST** services with **OpenTelemetry** distributed tracing, rate limiting, and circuit breakers on **AWS ECS/Kubernetes** sustaining 99.9% uptime for 10,000+ daily enterprise users.

Software Engineer II *Jul 2022 - Jun 2025*

Samay Systems LLC | B2B SaaS Compliance Platform

- Built **agent orchestration** systems using **CrewAI**, **LangChain**, and **LangGraph** for enterprise compliance automation, coordinating specialized agents for document ingestion, classification, and evidence summarization across 10K+ monthly records.
- Engineered **LangChain** RAG pipelines with **Azure AI Search** vector retrieval and **HuggingFace** embedding models, delivering natural language Q&A over enterprise compliance corpora at 89% retrieval precision for 15+ clients.
- Integrated **OpenAI API** for unstructured document summarization and auto-categorization into the core compliance product, reducing manual auditor effort by 70% and establishing the team's first LLM deployment.
- Built **Apache Kafka** event-driven microservices in **Python** and **Java** (Spring Boot) with dead-letter queues, idempotent consumers, and retry logic, decoupling 6+ services and improving LLM throughput by 50% at peak.
- Provisioned **Azure** infrastructure with **Terraform** and containerized LLM services with **Docker/Kubernetes**, **CI/CD** pipelines with versioning, deployments, and automated rollback, reducing AI deployment errors by 40%.
- Collaborated across ML research, platform engineering, and product teams to translate compliance domain requirements into agent architectures, shipping 4+ LLM-driven product features per quarter across the org.

Software Engineer I *Jul 2020 - Aug 2022*

Zemoso Technologies | LabOps Platform

- Built distributed **Java** Spring Boot backend services and **REST APIs** with **PostgreSQL** over a regulated multi-module scientific platform, handling experiment data tracking for 500+ scientists across 10+ modules.
- Developed full-stack features with **React.js** and **TypeScript** front-ends wired to async backend APIs, delivering responsive data entry and retrieval workflows adopted across all 10+ LabOps product modules.
- Designed normalized **PostgreSQL** schemas and **Redis**-cached query layers for high-read experiment data endpoints, improving retrieval performance by 30% and supporting concurrent access for 500+ users.
- Implemented **Spring Security** with OAuth2 role-based access control enforcing data isolation across research teams, and authored 100+ **JUnit** and **Jest** test cases that reduced production regression incidents by 35%.

Skills

Programming Languages: Python, TypeScript, JavaScript, Java, Go (basic), SQL
AI Agents & LLMs: LangGraph, LangChain, Pydantic AI, CrewAI, OpenAI API (GPT-4o), RAG, Prompt Engineering, Human-in-the-Loop, Memory Systems, Structured Outputs
MLOps & Evaluation: LLM evaluation pipelines (faithfulness, groundedness, latency), model versioning, canary
Vector Databases & Search: Pinecone, Azure AI Search, Semantic Search, Embedding Models
APIs & Backend: FastAPI, Node.js, Spring Boot, GraphQL, REST, Microservices, Apache Kafka, Event-driven Architecture
Cloud & DevOps: AWS (ECS, Lambda, SQS, S3, RDS), Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
Databases: PostgreSQL, Redis
Tools & Methodologies: Git, JIRA, Jest, JUnit, Agile/Scrum, OAuth2/JWT, Spring Security, System Design

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India